Predicting Destruction: Determinants of Violence in the 2002 Gujarat Riots*

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Abstract

This paper examines the relationship between social and demographic determinants from communities in Gujarat have for the intensity of inter-communal riots after an exogenous shock. In February 2002, a train bombing in Godhra, Gujarat, caused large riots lasting months throughout the state, with scholarship finding media, migration, and religious composition as potential factors for violence. Using demographic data from the 1991 and 2001 Indian census, along with data on the destruction of Muslim religious buildings, I construct a model that exploits demographic variation across districts to determine how social factors can determine riot intensity. Findings show that there are no statistically significant effects that factors of interest, language distribution, and religious migration have on riot intensity. However, all estimated coefficients have signs consistent with prior studies. These results suggest that there should be more empirical work done to understand the causes of the intensity of riots in Gujarat.

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1 Introduction

The 2002 Gujarat riots offers analysis on how the composition of a community can be a determinant of the destructiveness of inter-communal riots. Gujarat is unique because of the high magnitude of riots compared to other states in India. [17] In Gujarat, while some cities like Ahmedabad faced months of arson and vandalism, cities like Bhuj didn't receive the same impact. My question is: **What impact does the social and physical composition of a community have on the intensity of inter-communal riots following an exogenous shock incident in the Indian state of Gujarat?** I start with reviewing relevant literature about inter-communal riots with political and economic motivation for the 2002 riots in Gujarat. Then, I explain the process of constructing my data for analysis, which is used for my empirical model based on multiple linear regression using OLS. The results show there is not a statistically significant relationship between social and physical compositions of a community on intensity of riots.

1.1 Motivation

Gujarat is a unique state in India because of the number and intensity of riots that occur within its borders. Riots from Gujarat can serve as indicators of how tensions between Hindu, Muslim, Sikh, and other religious communities exist within broader South Asia. The state of Gujarat has important industries including manufacturing, pharmaceuticals, jewelry, and, as of recently, information technology. An estimated 60 million USD (570 Crore INR) was lost in damages directly related to riots. Over 1000 known people were killed from riots, but that number is considered a large underestimate [6]. Outside of Gujarat in 2002, the economic impacts of inter-communal violence in India amount to over 991.2 billion USD in 2019, which is over 75 of the combined military and internal security expenditure for the entire country [[19]. Along with economic loses, riots deter business firms from investing into the region. If a firm anticipates that a city or state has a high likelihood of riots and

economic losses, firms will exit or move elsewhere for business. Firm behavior in response to riots both directly and indirectly impact economic growth and prosperity in India.

2 Background

In this section I will explore the history of Hindu and Muslim inter-communal violence in India and Gujarat. I will summarize based on existing research, the political and economic factors why this type of violence in the region. Furthermore, I will review literature associated with previous inter-communal riot incidents throughout all of India and Gujarat, along with current scholarship on the 2002 Gujarat riots.

2.1 History of Inter-Communal Violence

Inter-communal incidents have existed in India since the time before British colonization. Gujarat has mythological foundations for both Hindus and Muslims. In Hinduism, Lord Krishna founded the Dvaraka Kingdom in current day Western Gujarat. In Islam, many Muslim warriors migrated from Ancient Persia and established Medieval Kingdoms that contributed to Gujarat's historical trade dominance. Along with those settlements, Sufi merchants and missionaries settled by the coast and served as a bridge for communal harmony among different religions [13]. However during and after British rule, authorities divided the population by emphasizing the difference between Hindus and Muslims, which promoted violence among groups [18] Creating deliberate violence allowed the British to exert harsher rule over the British Raj. From partition, the create of India and Pakistan furthered religious tensions by forcing the migration of millions of Muslims away from Gujarat to the Pakistani province of Sindh West [4]. Spouts of violence became more common as Muslims started to become a greater minority in the region.

In cities like Ahmedabad, Surat, and Vadodara, the ghettoization of Muslims into slums furthered disparities between Hindus and Muslims in Gujarat. Gujarati Muslims had fewer

economic opportunities and faced religious discrimination [8]. During the period from the 1970s to the 1990s, there were constant threats of riots across Gujarat, but they were often localized at the taluka (sub-district) or city level. While riots lessened in the new millennium, small incidents were occurring in different communities. In 2001, Gujarat recorded its deadliest earthquake in history. The 2001 Kutch earthquake in Gujarat led to over 20,000 deaths and thousands of buildings destroyed across Gujarat, but particularly felt in the western region [16]. The earthquake and preceding aftershocks led many to think that such a catastrophic event that affected people across socioeconomic status, religion, and caste could unify a divided people. However, that peace lasted less than a year.

2.2 2002 Godhra Train Burning

On February 27th 2002, a train of Hindu pilgrims from the holy city of Ayodhya returned to Gujarat on the Sabarmati Express. As the train left the platform, an emergency brake was pulled, stopping the train. Then a mob of around 2000 people stormed the train and attacked people inside. While this happened, a fire started inside 4 of the train cabins, trapping people inside. The cause of the fire remains under dispute. In total, 59 people died, and 48 were injured [1]. The mob that attacked the train were mainly Muslim. Because of an association with the mob with the entire Muslim community in Gujarat, the following riots were targeted towards the Gujarati Muslim community as a whole. This led to the destruction of Muslim businesses, Muslim religious sites, and generally areas were Muslims historically lived. These riots were different from past sectarian violence because of their scale and widespread effect across the whole state. Moreover, the government of Gujarat, headed by Narendra Modi (current Prime Minister of India), was complicit in not preventing violence from happening by allowing mobs into Muslim communities and delaying police presence during riots. [10]. Reports on the number of people killed range between 1000 and 2000, but that number is an understatement. Along with destruction, the riots fostered greater distrust against the Muslim community and led the government to create laws that segregated Muslims. Today,

Muslims in Gujarat face continued systemic discrimination because of the 2002 Godhra train burning.

2.3 Literature Review

The literature I will focus on in this paper focuses on the economic and political determinants of inter-communal riots between Muslims and Hindus in Gujarat and India broadly. Scholarship suggests that there are many factors that contribute to the intensity of inter-communal riots.

One factor that has an effect on the intensity of violence is with language distribution. India is extremely linguistically diverse, meaning language is an important factor in the development and division of communities. In the state of Assam in Northeast India, language use in communities, particularly with Bengali and Assamese were important factor in violence occurring [5]. This paper highlights how language causes friction between communities in the same state. In Gujarat this stratification is compounded with religion. For Hindus, most grow up learning Gujarati and use limited Hindi as a second language, while Muslim's first language is Hindi/Urdu, and they use Gujarati to get by. There is overlap between Hindu and Muslim speakers, but the difference is that Hindus are less likely to interact with non-Gujarat materials compared to Muslims, since Muslims have a greater understanding of Hindi/Urdu.

Related to language, the distribution of media was important in contributing to the intensity of riots. Saifuddin Ahmed explores the theory of media, particularly regional language papers, in creating non-object media. In his paper, he points to the Gujarat Samachar and Sandesh, the largest Gujarati language newspapers, as perpetrators of the 2002 riots [3]. A flagrant example of the biased nature of these outlets was when "Sandesh published photographs of the burning coach of the Sabarmati Express with the headline, 'Fifty Hindus burnt alive' above the masthead. Besides, it also had a gruesome colour spread of photographs of the Godhra corpses" [9]. During that time, non-Gujarati language

papers (including Hindi, English, Telugu, and more) were considered more objective about the events of 2002. People with only Gujarati knowledge would have higher exposure to propagandized media, whereas people with limited Gujarati usage are more likely to be exposed to objective media. Thereby, showing a motivation to commit violence.

Figure 1: Map of Gujarat with Muslim Monuments Destroyed

Looking specifically at the 2002 riots, a paper by Saumrita Jha emphasizes the importance of historical Islamic rule on riot intensity. In his analysis, he focused on historical data from Muslim ports and leaders before Partition. His theory is that districts with histories of Muslims leaders are more likely to exhibit religious harmony [11]. In his model, he considers factors of distance to Godhra, the BJP vote share, and demographic data on contributions for violence. He found that areas with medieval Islamic ports and were close to Godhra had statistically significant levels of violence. Jha's paper is useful since it shows how these riots were shaped by hundreds of years of history. Along with Jha, Chandhoke utilizes history to explain riot intensity. In her research, she specifically looks at the migration of Muslim communities within Gujarat before 2002 [8]. Starting in 1969, Muslims were forced into worse neighborhoods and barred from mixed communities. Then in 1991, the Gujarat government implemented the Disturbed Areas Act, which authorizes the state government to monitor or restrict property transactions between different religious communities in designated areas to prevent consolidation of holdings by one group [12]. This inflamed religious tensions

since the government would prevent Muslims from building in Hindu areas and furthered the ghettoization of Muslim communities. Chandhoke finds that these acts contributed to a mass migration of Muslims from rural and mixed areas of Gujarat into segregated urban areas. This migration started in 1991 and influenced violence in 2002, since Muslims could be easily identified based only on their neighborhood. Districts with high migration of Muslims could be targeted more by extremists [8].

The literature focuses on the factors for violence on a political level. Papers that use an empirical model to analyze the destruction of the 2002 riots focus on demographic data and historical politics before 2002. There is no empirical analysis for language distribution in Gujarat and the impact of migration on riots. I intend to fill this gap by providing an empirical approach to investigating whether the political theories for riot intensity hold among the data. By researching language distribution and migration patterns as causes for riots on an empirical level, I position myself in scholarship by providing evidence that can support these theories.

3 Data Set

This section discusses the data involved with creating my analysis. There will be explanations on the create of data and rationale for including and excluding certain variables. There are three different types of variables used for this analysis. **Total Muslim Monuments Destroyed** is the dependent variable I am investigating. **Percent Muslim Change 1991-2001**, **Percent Gujarat Mother Tongue** are my principal independent variables, while **Percent Hindu Minus Muslim** and **Good Building Stock** are my main control variables.

3.1 Indian Census Data (1991 and 2001)

The majority of my data comes from the 2001 Indian census [15]. Every 10 years, India conducts one of the largest censuses in the world. In the census they catalog conventional demographic data like population, district/city size, but also are mandated to collect data on religion, caste, and mother tongue data, which is not seen in the United States. Because the riots occurred in 2002, I wanted to use recent demographic and spatial data that would capture variation between districts in Gujarat. I am assuming that there are no significant demographic changes between the 2001 census are population in 2002. Also, since the 2001 census was conducted after the 2001 earthquake, the loss of life and damages related to infrastructure were taken into account within census data, particularly of districts in Western Gujarat. My 2001 census data is focused on areas counted as "non-rural"; this means I am excluding areas in Gujarat that don't have formal settlements, including the tribal areas home to Adivasi communities. This is not a problem in my data since Muslims are counted entirely from the towns and city data within each district.

I use the 1991 census to show the percentage change of Muslims within a district [14]. Because of the 1991 implementation of the Disturbed Areas Act, the 10-year period from 1991 to 2001 shows migration that occurred thereafter. The change in percentage is meant to show historic migration preceding the riots. Otherwise, I utilize linguistic data about mother tongue from 2001, since Gujarati as a mother tongue can indicate exposure to propaganda for people within districts. Using the difference between Hindus and Muslims within a district shows the extent to which Muslims are a minority in a district.

3.2 Destruction of Muslim Monuments

The impact of the riots hit all aspects of life for the Gujarati Muslim community. The riots led to the death of over 1000 people, but counts are extremely unreliable since many agencies do not account who died in pogroms. Accounts for deaths are literally changing

every year because there is an active effort to minimize the impact of these riots. Even using archival data does not provide an accurate picture of the human destruction from the riots. Instead of focusing on data relating to deaths, I choose to look at desecration of Muslim religious monuments. I use data from the Center for Justice and Peace (CJP), which is a non-government human rights organization that catalogs data from events that other government agencies do not properly cover. For this analysis I am using their data from 2002 about religious monuments across Gujarat [7]. I am combining 3 different Muslim religious monuments that signal violence against Muslim communities. I add **Masjids** (Mosques), **Darghas** (Sufi Tombs), and **Madrasas** (historical Islamic schools) together to create **Total Muslim Monuments Destroyed** Figure 4.

3.3 Limitations

When working with this data, there are limitations on how I am able to properly construct a dataset without losing significant portions of my data. The largest problem is that monuments don't capture the true impact of inter-communal riots compared with the casualties. A building can be rebuilt, but someone's life can never be taken back. Also, due to time constraints and difficulty attaining data on casualties, I elected to use monument destruction as an outcome for riot violence. Another issue is that my sample size is extremely small, since I am looking at 23 and 24 districts. I did not have sufficient data to count the talukas subdistrict. Furthermore, the district of Sabarkantha is not included since the data from 1991 does not exist. Since I take data from the 1991 and 2001 censuses, there are different numbers of districts from that time, so I had to manually insert data to accurately represent each district. If I were to continue with this project, I would find better data that reflected sub-district levels in Gujarat.

4 Empirical Framework

In this section, I show my empirical model and explain my rationale behind constructing my independent terms. Since this model is static, there won't be significant discussion on model complexity; however, I will go over each variable used.

Regression Specification

$$\begin{aligned} MuslimDestruction_i = \beta_0 &+ \beta_1\, pctGujarati_i + \beta_2\, pctMuslimChange_i \\ &+ \beta_3 (pctHinduMinusMuslim_i)^2 \\ &+ \beta_4\, goodHousing_i \\ &+ \epsilon_i \end{aligned}$$

where:

$MuslimDestruction_i$	Muslim sites destroyed per 100k population in district i
$pctGujarati_i$	Share of Gujarati mother-tongue speakers, 2001
$pctMuslimChange_i$	Change in urban Muslim population share, 1991–2001
$(pctHinduMinusMuslim_i)^2$	Squared Hindu–Muslim population share difference
$goodHousing_i$	Good-condition housing units per capita

4.1 Rationale

I estimate the model using OLS because the 2002 riots in Gujarat can be treated as a large, exogenous shock affecting districts across the state. My dependent variable captures the total level of Muslim monument destruction associated with this single episode of violence.

Because the outcome reflects the cumulative effects of one major event rather than a process unfolding over many years, I do not exploit time variation and instead rely on cross-district variation. While constructing a time-varying dataset on riots or monument destruction would allow for panel methods, such data were not available for this project. Future work could extend the analysis using time-series or panel data on communal violence.

Since I am exploring migration and linguistic composition, I construct the dataset in a stepwise manner centered on those main explanatory variables. I first estimate specifications including these variables, and then progressively add control variables to examine whether additional district characteristics account for the observed variation in destruction across districts.

I make the difference of of Hindu and Muslims as a non-linear term because of a theory in political economy called "minority threat theory". The theory posits that the presence of a minority group does not have a linear relationship. Once a minority group, in this case Muslims, has an increased presence in a district, it is theorized that the majority group would become threatened and therefore increase violence.

5 Results

In this section I will go through and interpret two different regression tables. The first regression table is my independent variables of interest I am investigating without controls. Then, my second regression table is my primary independent variable while building in my controls.

5.1 Discussion

Table 1 highlights the independent variables I am investigating without the use of controls. When regressing separately and together, Percent Gujarati speakers or Percent Muslim Changed have a statistically significant effect on Muslim Monuments Destroyed. Although

Table 1: Stepwise OLS Regression Results without Controls
Dependent Variable: Muslim Religious Monuments Destroyed per 100,000

	Model 1 *Gujarati Share*	**Model 2** *Muslim Pop. Change*	**Model 3** *Full Model*
Intercept	−3.881	4.507**	−5.366
	(7.883)	(1.347)	(8.418)
% Gujarati	0.097		0.118
	(0.092)		(0.100)
% Muslim Change		0.079	0.128
		(0.180)	(0.183)
N	24	23	23
R^2	0.048	0.009	0.074
F-statistic	1.112	0.192	0.803
p-value (F)	0.3032	0.666	0.462

Note: Standard errors in parentheses. Models estimated via OLS. Model 2 and Model 3 exclude one observation due to missingness in % Muslim Change.
$^{*}p < 0.05$, $^{**}p < 0.01$, $^{***}p < 0.001$

Table 2: Stepwise OLS Regression Results with Controls
Dependent Variable: Muslim Religious Monuments Destroyed per 100,000

	Model 1	**Model 2**	**Model 3**	**Model 4**	**Model 5**
Intercept	−0.338	6.475	−0.222	−4.716	0.656
	(7.897)	(4.169)	(8.676)	(8.485)	(8.700)
% Gujarati	0.092		0.121	0.080	0.079
	(0.091)		(0.096)	(0.109)	(0.104)
% Muslim Change		0.012	0.096	0.090	0.052
		(0.172)	(0.177)	(0.189)	(0.181)
(Hindu − Muslim)²	−0.001†	−0.001	−0.001		−0.001
	(0.001)	(0.001)	(0.001)		(0.001)
Good Housing per Capita	5.424	8.995		6.402	6.988
	(5.709)	(6.129)		(7.042)	(6.730)
N	24	23	23	23	23
R^2	0.209	0.210	0.189	0.113	0.235
F-statistic	1.763	1.680	1.472	0.806	1.378
p-value (F)	0.187	0.205	0.254	0.506	0.281

Note: Standard errors in parentheses. Models estimated via OLS. Models 2–5 exclude one observation due to missingness in % Muslim Change.
$^{\dagger}p < 0.10$, $^{*}p < 0.05$, $^{**}p < 0.01$, $^{***}p < 0.001$

the full model, including both Percent Gujarat and Percent Muslim change, has the highest R^2 score, only 7.4pct of the variation is explained by those variables. The simple linear regression model with only the percent share of Gujarati speakers has the lowest p-value, but remains at levels of statistical significance. The results in Table 1 suggest that the independent variables of investigation alone do not meaningfully explain variation in destruction in 2002.

Table 2 introduces control variables to account for cross-district heterogeneity. Across all models, including control variables, there are higher R^2 compared to models in Table 1, indicating that a greater share of the variation in destruction is explained once observable district characteristics are included. However, none of the models are statistically significant below the $p < 0.10$. Despite this, the estimated coefficients are economically significant and follow theory. There are small positive associations between Percent Gujarat and Percent Muslim Change, which is consistent with theories linking Gujarati segmentation and Muslim migration to increased inter-communal tensions. But these results do not provide statistically strong evidence in support of these mechanisms.

5.2 Robustness Check

To assess the robustness of the model, I utilize diagnostic measures including Cook's distance and leverage. In RStudio, diagnostic plots can be generated to identify influential observations and potential outliers. Specifically, I examine the Cook's distance plot and the Residuals vs. Leverage plot. In both diagnostics, District 2 emerges as a clear outlier; this district corresponds to the Anand district.

Cook's distance is used to estimate the influence of an observation on the overall regression model:

$$D_i = \frac{e_i^2}{p\,MSE}\left(\frac{h_{ii}}{(1-h_{ii})^2}\right)$$

Values of Cook's distance above 0.5 are often interpreted as indicating potentially influen-

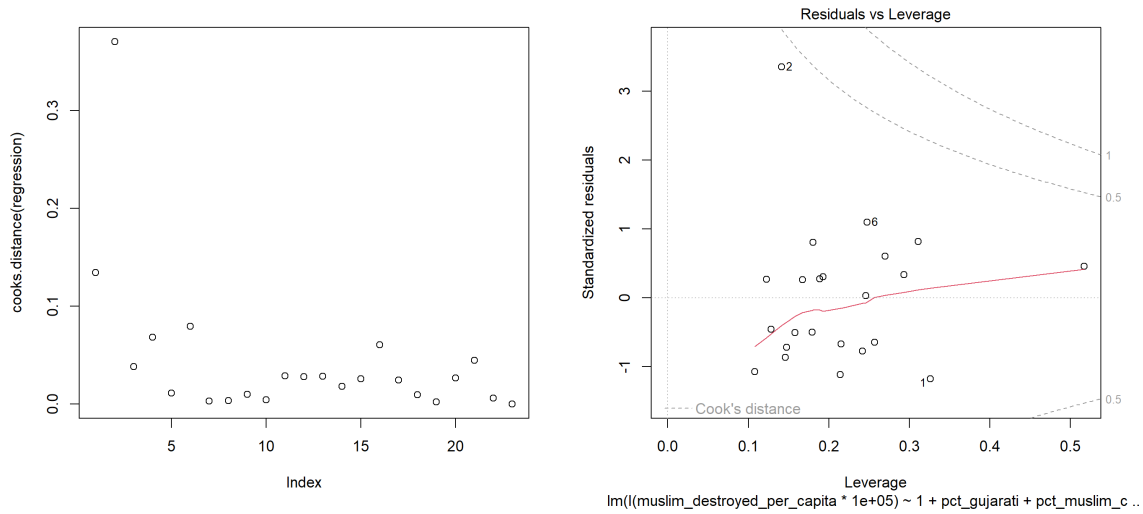

Figure 2: Cook's Distance Plot and Residuals vs. Leverage Plot for the Robustness Check

tial observations, while values approaching or exceeding 1 suggest highly influential points. The Cook's distance for the Anand district is 0.3706, indicating that the observation has a moderate influence on the regression, though it does not exceed conventional thresholds for a highly influential outlier. When excluding Anand, the models have a lower R^2 and p-value, which could mean that Anand District acted as an influential point that inflated the observed effect size. I run the same regression, but exclude the Anand district, which is seen through Table 4 and Table 5 in the appendix.

There are several plausible explanations for why Anand appears as an outlier relative to other districts in the dataset. First, Anand is geographically situated between Ahmedabad and Vadodara, both of which are highly urbanized districts with large populations and significant political and economic influence. This location may expose Anand to spillover effects from surrounding urban centers, including higher levels of inter-district mobility to conflict dynamics. As a result, individuals from neighboring districts may have stronger incentives to target areas outside their immediate communities, potentially increasing observed levels of destruction in border districts such as Anand. Second, Anand has a relatively high proportion of Non-Resident Indians (NRIs) compared to other districts [2]. NRIs are individuals of Indian origin who reside outside India and do not hold Indian citizenship, as India does

not permit dual citizenship. One possible interpretation is that districts with higher NRI populations may exhibit different social dynamics in a community. Someone who has the mobility to attain a foreign citizenship are less likely to be invested in a community, so they could have a higher incentive to desecrate monuments. However, the relationship between the NRI population is likely indirect and determined by broader socioeconomic factors.

5.3 Limitations

There are many limitations with the results of this project. As mentioned before, using religious monuments destroyed doesn't fully account for the destruction faced in a community by riots. With the results, the percentage of Gujarati speakers as a proxy for media becomes complicated when it doesn't necessarily account for speakers of multiple languages and Muslims who mainly speak Gujarati. This could account for the result not being statistically significant, since there are unobserved variations in mother tongue speakers of Gujarati not captured by the variable. The models that include the difference of Hindus minus Muslims squared absorbs variation within the model, which could come from areas with higher Muslim populations have a higher concentration of religious monuments, therefore are more exposed to being destroyed.

5.4 Implications

Literature on inter-communal riots in South Asian explains that social factors such as language distribution, religious composition, and migration would contribute to widespread inter-communal riots like those seen in 2002. However this paper shows that there is little statistical evidence that this could be the case. These findings suggest that more empirical work should be done to determine of there is a causal relationship between language composition and migration of religious minorities on the formation of riots. There appears to be limited economic significance in the relationship between language composition and migration as drivers of destruction, although the estimated coefficients have signs consistent

with theoretical expectations. While the results are not statistically significant, the positive direction of the estimates suggests that the proposed mechanisms may still play a role. Future research should expand data collection and further investigate these relationships in greater depth. Additionally, it is important to note the barriers researchers face when studying controversial topics like the 2002 Godhra riots, which can make reliable data collection and analysis particularly challenging.

6 Conclusion

This study investigates the determinants of inter-communal violence in Gujarat by exploiting exogenous nature of the 2002 Godhra riots. This study utilizes census data to investigate how language distribution and migration contribute to riot intensity in districts. The paper reveals that there is no statistical significance regarding the variables of interest for contributing to riot intensity. There is limited economic significance since the coefficients follow the signs related to the theory on riot determinants. These results emphasize the importance of continuing empirical research on inter-communal riots, specifically pertaining to the 2002 riots in Gujarat.

Further research should focus on the role that media distribution and the government of Gujarat played in inciting and influencing riots. Data should be collected from the Taluka level to capture variation within districts, since the districts have large populations and areas. Moreover, further studies should examine economic variation within communities that can contribute to riot intensity. This would provide a more comprehensive understanding of how action can be taken to reduce intense riots from occurring.

It is important to continue researching the 2002 riots in Godhra. There is an active effort to hide the history that is necessary for understanding Hindu-Muslim relations in modern South Asia. While these findings don't have statistical significance, they nevertheless provide valuable insights into the structural and contextual factors associated with riot intensity. Inter-communal riots are inherently difficult to predict because of the unpredictability that

comes with human behavior. By taking a granular approach to create insights on why intensity is vastly different, it allows actionable policy interventions to prevent the loss of life and property seen in 2002.

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**I disclose that Claude LLM assisted me with data cleaning, Latex troubleshooting, debugging, and grammar choice. If you have questions or concerns regarding my use of Artificial Intelligence, please reach out to me on email rchinnakotla@wisc.edu*

6.1 Github Repository

`https://github.com/rchinnakotla131/ECON580`

7 Appendix

Table 3: Muslim Monuments Destroyed by District, Gujarat

#	District	Count	Per 100,000
1	Amreli	0	0.000
2	Anand	74	26.860
3	Bhavnagar	5	4.010
4	Banaskantha	0	0.000
5	Bharuch	7	0.511
6	Gandhinagar	23	4.922
7	Ahmedabad	104	2.284
8	Dahod	20	4.970
9	Junagadh	14	0.967
10	Kheda	65	7.773
11	Jamnagar	0	0.000
12	Kutch-Bhuj	4	0.819
13	Mehsana	30	9.586
14	Narmada	1	0.107
15	Navsari	0	0.000
16	Godhra	48	11.810
17	Patan	12	5.745
18	Surat	13	8.316
19	Vadodara	42	2.551
20	Porbandar	4	7.677
21	Rajkot	11	0.367
22	Sabarkantha	82	3.936
23	Surendranagar	1	0.359
24	Valsad	1	0.354

Note: "Per 100,000" expresses the number of destroyed properties per 100,000 residents in each district.

Table 4: Robustness Check: Stepwise OLS Regression Results without Controls (Anand Excluded)

Dependent Variable: Muslim Religious Monuments Destroyed per 100,000

	Model 1 *Gujarati Share*	**Model 2** *Muslim Pop. Change*	**Model 3** *Full Model*
Intercept	-1.709	3.374^{***}	-2.233
	(4.799)	(0.844)	(5.194)
% Gujarati	0.060		0.068
	(0.056)		(0.062)
% Muslim Change		0.023	0.053
		(0.110)	(0.113)
N	23	22	22
R^2	0.051	0.002	0.061
F-statistic	1.14	0.044	0.621
p-value (F)	0.300	0.835	0.548

Note: Standard errors in parentheses. Models estimated via OLS. Anand (dist_code = 2) was excluded as an outlier observation. Model 2 and Model 3 exclude one additional observation due to missingness in % Muslim Change. $^{*}p < 0.05$, $^{**}p < 0.01$, $^{***}p < 0.001$

Table 5: Robustness Check: Stepwise OLS Regression Results with Controls (Anand Excluded)

Dependent Variable: Muslim Religious Monuments Destroyed per 100,000

	Model 1	**Model 2**	**Model 3**	**Model 4**	**Model 5**
Intercept	1.572	$5.379^{\dagger}$	0.715	−2.124	0.954
	(4.900)	(2.654)	(5.341)	(5.333)	(5.484)
% Gujarati	0.052		0.072	0.060	0.060
	(0.057)		(0.060)	(0.068)	(0.065)
% Muslim Change		−0.006	0.037	0.045	0.025
		(0.109)	(0.109)	(0.118)	(0.114)
(Hindu − Muslim)²	−0.001	−0.001	−0.001		−0.001
	(0.000)	(0.000)	(0.000)		(0.000)
Good Housing per Capita	1.110	3.521		1.414	2.042
	(3.654)	(4.020)		(4.500)	(4.343)
N	23	22	22	22	22
R^2	0.166	0.145	0.175	0.066	0.186
F-statistic	1.261	1.016	1.273	0.427	0.969
p-value (F)	0.316	0.409	0.314	0.736	0.450

Note: Standard errors in parentheses. Models estimated via OLS. Anand (dist_code = 2) was excluded as an outlier observation. Models 2–5 exclude one additional observation due to missingness in % Muslim Change.

$^{\dagger}p < 0.10$, $^{*}p < 0.05$, $^{**}p < 0.01$, $^{***}p < 0.001$

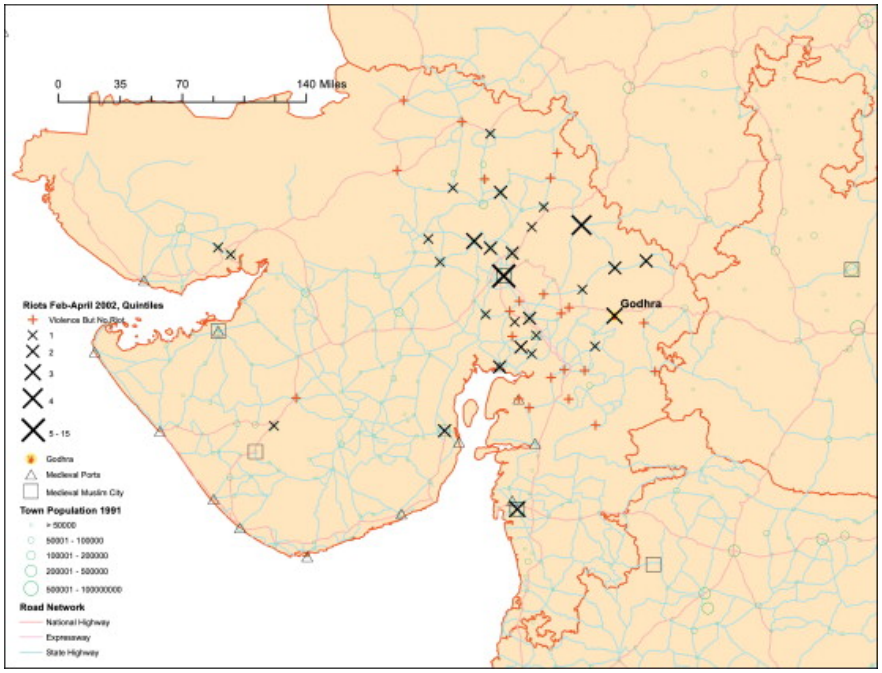

Figure 3: Communal violence in Gujarat (adapted from Jha [11]).

Table 6: District-Level Gujarati Language Share and Muslim Population Change

#	District	% Gujarati	% Muslim Change
1	Amreli	58.271	+2.870
2	Anand	91.564	+0.789
3	Bhavnagar	97.249	+0.200
4	Banaskantha	94.694	+0.250
5	Bharuch	84.970	−9.796
6	Gandhinagar	78.600	+1.591
7	Ahmedabad	81.103	+0.006
8	Dahod	97.714	−13.944
9	Junagadh	96.355	−7.138
10	Kheda	81.180	+0.789
11	Jamnagar	93.458	−13.772
12	Kutch-Bhuj	92.239	+3.252
13	Mehsana	97.652	+4.784
14	Narmada	92.562	−9.796
15	Navsari	90.470	+5.149
16	Godhra	93.419	−13.944
17	Patan	83.601	+0.250
18	Surat	68.698	+12.926
19	Vadodara	77.817	+0.148
20	Porbandar	93.705	−7.138
21	Rajkot	50.423	−12.078
22	Sabarkantha	97.850	—
23	Surendranagar	82.775	+5.149
24	Valsad	59.733	+2.616

Note: % Gujarati is the share of the district population reporting Gujarati as their mother tongue. % Muslim Change is the percentage point change in the urban Muslim population share between 1991 and 2001. Sabarkantha is missing Muslim change data due to boundary changes between census years.

(a) Dargah of Mai Saheba, Dandi, Gujarat. Wikimedia Commons.

(b) Pir Ghazi Madrasa, Gujarat. Wikipedia.

(c) Sahal Ki Masjid, Champaner, Gujarat. Wikimedia Commons.

Figure 4: Islamic religious sites in Gujarat